

Summary of Previous Work

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In this article, we summarize results in [2, 4], that are necessary or useful for implementing a theorem proving method in [3]. Moreover, we show techniques used in a prototype of the implementation, that are not described in [3].

Let $\mathcal{F}$ and $\mathcal{G}$ be sets of function symbols such that $\mathcal{F} \cap \mathcal{G} = \emptyset$, and $\mathcal{P}$ be a set of predicate symbols, and $\mathcal{M}$ be a structure that gives interpretation of symbols in $\mathcal{G}$ and $\mathcal{P}$. Let $\mathcal{R}$ be a constrained TRS over $\mathcal{F}$, $\mathcal{G}$, $\mathcal{P}$, $\mathcal{M}$, and $\mathcal{V}$.

1 Summary of [2]

1.1 Transformation of Procedural Programs into Constrained TRSs

In [2], a transformation from functions over integers written in C language into constrained TRSs has been proposed, while the transformation is natural and similar ones can be seen in other papers.

Example 1 Consider the following program:

```
int sum1(int n){  
  int i = 1;  
  int z = 0;  
  while( i <= n ){  
    z = z + i;  
    i = i + 1;  
  }  
  return z;  
}
```

This program is transformed into the following constrained TRS:

$$\mathcal{R}_{\text{sum1}} = \left\{ \begin{array}{ll} \text{sum1}(n) \rightarrow \text{u}_1(n, \text{s}(0)) \\ \text{u}_1(n, i) \rightarrow \text{u}_2(n, i, 0) \\ \text{u}_2(n, i, z) \rightarrow \text{u}_3(n, i, z) & \llbracket i \leq n \rrbracket \\ \text{u}_3(n, i, z) \rightarrow \text{u}_4(n, i, \text{add}(z, i)) \\ \text{u}_4(n, i, z) \rightarrow \text{u}_5(n, \text{add}(i, \text{s}(0)), z) \\ \text{u}_5(n, i, z) \rightarrow \text{u}_2(n, i, z) \\ \text{u}_2(n, i, z) \rightarrow \text{u}_6(n, i, z) & \llbracket \neg i \leq n \rrbracket \\ \text{u}_6(n, i, z) \rightarrow \text{u}_7(z) \\ \text{u}_7(y) \rightarrow y \end{array} \right\} \cup \mathcal{R}_{PA}$$

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where the universe of the structure $\mathcal{M}$ is $\mathbb{Z}$, $s^{\mathcal{M}}(x) = x + 1$, $p^{\mathcal{M}}(x) = x - 1$, $0^{\mathcal{M}} = 0$, $\leq^{\mathcal{M}}(x, y) = x \leq y$, and

$$\mathcal{R}_{PA} = \left\{ \begin{array}{ll} \text{add}(0, y) & \rightarrow y \\ \text{add}(s(x), y) & \rightarrow s(\text{add}(x, y)) \\ \text{add}(p(x), y) & \rightarrow p(\text{add}(x, y)) \\ s(p(x)) & \rightarrow x \\ p(s(x)) & \rightarrow x \end{array} \right\}$$

$\mathcal{R}_{\text{sum1}}$ is optimized as follows:

$$\left\{ \begin{array}{ll} \text{sum1}(n) & \rightarrow u_2(n, s(0)) \\ u_2(n, i, z) & \rightarrow u_2(n, \text{add}(i, s(0)), \text{add}(z, i)) \quad \llbracket i \leq n \rrbracket \\ u_2(n, i, z) & \rightarrow z \quad \llbracket \neg i \leq n \rrbracket \end{array} \right\} \cup \mathcal{R}_{PA}$$

A similar transformation can be found in [1].

1.2 Confluence of Constrained TRSs

Let ϕ_1 and ρ_2 be constrained rewrite rules in $\mathcal{R}$, and $l_1 \rightarrow r_1 \llbracket \phi_1 \rrbracket$ and $l_2 \rightarrow r_2 \llbracket \phi_2 \rrbracket$ be variants of ρ_1 and ρ_2 , resp., such that $\text{Var}(l_1) \cap \text{Var}(l_2) = \emptyset$. Suppose that l_1 is of the form $C[t]$ for some context $C[\]$ and some non-variable term t ($\notin \mathcal{V}$) such that t and l_2 be unifiable, where if ρ_1 and ρ_2 are the same then $C[\] \not\equiv \square$. Given a most general unifier σ of t and l_2 , the constrained pair $(r_1\sigma, (C[r_2])\sigma) \llbracket (\phi_1 \wedge \phi_2)\sigma \rrbracket$ is called a *constrained critical pair*. Two constrained rewrite rules are said to be *overlapping* (or, in the case of ρ_1 and ρ_2 above, ρ_2 overlaps with ρ_1 at position p to the hole $\square$) if there exists a critical pair between the two rules. We denote the set of critical pairs between rules ρ_1 and ρ_2 by $CP(\rho_1, \rho_2)$. The set of critical pairs between ρ_1 and arbitrary rules in $\mathcal{R}$ is denoted by $CP(\rho_1, \mathcal{R})$: $CP(\rho_1, \mathcal{R}) = \bigcup_{\rho_2 \in \mathcal{R}} CP(\rho_1, \rho_2)$. The set of critical pairs between arbitrary rules in $\mathcal{R}$ is denoted by $CP(\mathcal{R})$: $CP(\mathcal{R}) = \bigcup_{\rho_1 \in \mathcal{R}} CP(\rho_1, \mathcal{R})$.

A critical pair $(s, t) \llbracket \phi \rrbracket$ is called *feasible* if ϕ is satisfiable w.r.t. $\mathcal{M}$. The critical pair is called *infeasible* if it is not feasible. We consider two rules to be non-overlapping if there exists no feasible critical pairs between the two rules.

Terms s and t are called *joinable under a constraint ϕ* if $s\sigma \downarrow_{\mathcal{R}} t\sigma$ for any substitution σ such that $\phi\sigma$ is valid w.r.t. $\mathcal{M}$. A critical pair $(s, t) \llbracket \phi \rrbracket$ is called *joinable* if s and t are joinable under ϕ .

In general, Critical Pair Lemma for TRSs does not hold for constrained TRSs (cf. Mitschke's theorem in λ -calculus).

Definition 2 ([2, Definition 5.1]) *Let $\rightarrow$ be a binary relation on terms. We say that $\rightarrow$ preserves interpreted terms if for any interpretable terms $t \in T(\mathcal{G}, \mathcal{V})$, there is no term u in $T(\mathcal{F} \cup \mathcal{G}, \mathcal{V}) \setminus T(\mathcal{G}, \mathcal{V})$ such that $t \rightarrow^* u$. For a binary relation $\rightarrow$ preserving interpreted terms, a constraint ϕ is called closed under $\rightarrow$ if for any substitution σ with $\text{Dom}(\sigma) \subseteq \text{fv}(\phi)$, there is no substitution θ such that*

- $x\sigma \rightarrow^* x\theta$ for any $x \in \text{fv}(\phi)$, and
- $\phi\sigma \Leftrightarrow \phi\theta$ is not valid w.r.t. $\mathcal{M}$.

$\mathcal{R}$ is called *stably-constrained w.r.t. $\rightarrow$* if for every constrained rule $l \rightarrow r \llbracket \phi \rrbracket$ in $\mathcal{R}$, the constraint ϕ is stable w.r.t. $\rightarrow$. Especially, $\mathcal{R}$ is called *stably-constrained* if $\rightarrow_{\mathcal{R}}$ preserves interpreted terms and $\mathcal{R}$ is stably-constrained w.r.t. $\rightarrow_{\mathcal{R}}$.

Proposition 3 $\mathcal{R}$ is stably-constrained if $\mathcal{R}$ is locally sound w.r.t. $\mathcal{M}$.

Lemma 4 ([2, Lemma 5.3]) Let $\mathcal{R}$ be a stably-constrained TRS, and s, t_1, t_2 be terms such that $s \rightarrow_{\mathcal{R}} t_1$ and $s \rightarrow_{\mathcal{R}} t_2$. Then, one of the following holds:

- $t_1 \downarrow_{\mathcal{R}} t_2$, or
- $t_1 \equiv s[u_1]_p$ and $t_2 \equiv s[u_2]_p$ where there exists a critical pair $(v_1, v_2) \llbracket \phi \rrbracket$ in $CP(\mathcal{R})$ and substitution σ such that $\phi\sigma$ is valid w.r.t. $\mathcal{M}$, and
 - $u_1 \equiv v_1\sigma$ and $u_2 \equiv v_2\sigma$, or
 - $u_1 \equiv v_2\sigma$ and $u_2 \equiv v_1\sigma$.

Theorem 5 ([2, Theorem 5.4]) Let $\mathcal{R}$ be a stably-constrained TRS. $\mathcal{R}$ is locally confluent iff every feasible critical pair in $CP(\mathcal{R})$ is joinable.

Corollary 6 A terminating stably-constrained TRS $\mathcal{R}$ is confluent iff every feasible critical pair in $CP(\mathcal{R})$ is joinable.

1.3 Reduction Under Constraints

In procedures of *completion* and *rewriting induction*, both sides of constrained equations or the right-hands sides of constrained rewrite rules are reduced. In such a case, the constraint part performs as an environment. For example, $f(x) \approx x \llbracket x > 0 \rrbracket$ means that the equivalence relation $f(x) \approx x$ holds if $x > 0$. In this case, we can reduce the left-hand side $f(x)$ by $\{f(x) \rightarrow x \llbracket x \geq 0 \rrbracket\}$ since the equation is considered under the constraint $x > 0$. For a satisfiable constraint ψ , the *reduction $\rightarrow_{\mathcal{R}, \psi}$ under ψ* is formalized as follows:

$$\rightarrow_{\mathcal{R}, \psi} = \{(C[l\sigma], C[r\sigma]) \mid l \rightarrow r \llbracket \phi \rrbracket \in \mathcal{R}, \psi \Rightarrow \phi\sigma \text{ is valid w.r.t. } \mathcal{M}\}$$

Proposition 7 ([2, Proposition 6.1]) $\mathcal{R}$ is terminating iff $\rightarrow_{\mathcal{R}, \psi}$ is terminating for every satisfiable constraint ψ .

1.4 Completion of Constrained TRSs

Inference rules of *completion* for TRSs are adapted to constrained TRSs as follows:

Expansion

$$\frac{(\mathcal{E} \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{R})}{(\mathcal{E} \cup CP(s \rightarrow t \llbracket \phi \rrbracket), \mathcal{R} \cup \{s \rightarrow t \llbracket \phi \rrbracket\}, \mathcal{R} \cup \{s \rightarrow t \llbracket \phi \rrbracket\})}$$

where $s \succ t$, $root(s) \notin \mathcal{G}$, and $fv(\phi) \subseteq Var(s)$.

Composition

$$\frac{(\mathcal{E}, \mathcal{R} \cup \{s \rightarrow t \llbracket \phi \rrbracket\})}{(\mathcal{E}, \mathcal{R} \cup \{s \rightarrow u \llbracket \phi \rrbracket\})}$$

where $t \rightarrow_{\mathcal{R}, \phi} u$.

Simplification

$$\frac{(\mathcal{E} \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{R})}{(\mathcal{E} \cup \{s \approx u \llbracket \phi \rrbracket\}, \mathcal{R})}$$

where $t \rightarrow_{\mathcal{R}, \phi} u$.

Deletion

$$\frac{(\mathcal{E} \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{R})}{(\mathcal{E}, \mathcal{R})}$$

where $s \equiv t$ or ϕ is not satisfiable w.r.t. $\mathcal{M}$.

Moreover, the following two inference rules are added:

E-Decomposition

$$\frac{(\mathcal{E} \cup \{s \simeq C[u\sigma] \llbracket \phi \rrbracket\}, \mathcal{R})}{(\mathcal{E} \cup \{s \approx C[v\sigma] \llbracket \phi \wedge \psi\sigma \rrbracket, s \approx C[u\sigma] \llbracket \phi \wedge \neg\psi\sigma \rrbracket\}, \mathcal{R})}$$

where $u \rightarrow v \llbracket \psi \rrbracket \in \mathcal{R}$, $\phi \Rightarrow \psi\sigma$ is valid w.r.t. $\mathcal{M}$, and $fv(\psi\sigma) \subseteq fv(\phi)$.

MGU-Deletion

$$\frac{(\mathcal{E} \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{R})}{(\mathcal{E} \cup \{s \approx t \llbracket \phi \wedge \neg \left(\bigwedge_{x \in \text{Dom}(\sigma)} x = x\sigma \right) \rrbracket\}, \mathcal{R})}$$

where $\mathcal{G}$, $\mathcal{P}$, and $\mathcal{M}$ correspond to linear integer arithmetic constraints, σ is a most general unifier of s and t , $\text{Dom}(\sigma) \subseteq fv(\phi)$, and $\text{Ran}(\sigma) \subseteq T(\mathcal{G}, fv(\phi))$.

When we apply one of the inference rules above to $(\mathcal{E}, \mathcal{R})$ and obtain $(\mathcal{E}', \mathcal{R}')$, we write $(\mathcal{E}, \mathcal{R}) \vdash_c (\mathcal{E}', \mathcal{R}')$.

Theorem 8 ([2, Theorem 6.4]) *Let $\succ$ be a reduction order over terms, $\mathcal{E}_0$ be a finite set of constrained equations, and $\mathcal{R}_0$ be a stably-constrained TRS such that $\rightarrow_{\mathcal{R}_0} \subseteq \succ$. Suppose that*

- $\mathcal{R}_0$ is complete w.r.t. ground terms, and
- for every constrained equation $s \approx t \llbracket \phi \rrbracket$ in $\mathcal{E}_0$, ϕ is stable w.r.t. $\rightarrow_{\mathcal{R}_0}$.

A derivation sequence $(\mathcal{E}_0, \mathcal{R}_0) \vdash_c \dots \vdash_c (\emptyset, \mathcal{R}_n)$ satisfies all of the following:

- $\mathcal{R}_n$ is equivalent to $\mathcal{E}_0 \cup \mathcal{R}_0$ w.r.t. ground terms: $\leftrightarrow_{\mathcal{R}_n}^* = \leftrightarrow_{\mathcal{E}_0 \cup \mathcal{R}_0}^*$,
- $\mathcal{R}_n$ is stably-constrained,
- $\mathcal{R}_n$ is complete w.r.t. ground terms, and
- $T(\mathcal{C}_{\mathcal{R}_0}, \mathcal{V}) = T(\mathcal{C}_{\mathcal{R}_n}, \mathcal{V})$ (in other words, $\mathcal{C}_{\mathcal{R}_0} = \mathcal{C}_{\mathcal{R}_n}$).¹

Theorem 9 ([2, Theorem 6.4]) *Let $\succ$ be a reduction order over terms, $\mathcal{E}_0$ be a finite set of constrained equations, and $\mathcal{R}_0$ be a stably-constrained TRS such that $\rightarrow_{\mathcal{R}_0} \subseteq \succ$. Suppose that*

¹This looks wrong.

- $\mathcal{R}_0$ is complete, and
- for every constrained equation $s \approx t \llbracket \phi \rrbracket$ in $\mathcal{E}_0$, ϕ is stable w.r.t. $\rightarrow_{\mathcal{R}_0}$.

A derivation sequence $(\mathcal{E}_0, \mathcal{R}_0) \vdash_c \dots \vdash_c (\emptyset, \mathcal{R}_n)$ without *E-Decomposition* or *MGU-Deletion* satisfies all of the following:

- $\mathcal{R}_n$ is equivalent to $\mathcal{E}_0 \cup \mathcal{R}_0$: $\leftrightarrow_{\mathcal{R}_n}^* = \leftrightarrow_{\mathcal{E}_0 \cup \mathcal{R}_0}^*$,
- $\mathcal{R}_n$ is stably-constrained,
- $\mathcal{R}_n$ is complete, and
- $T(\mathcal{C}_{\mathcal{R}_0}, \mathcal{V}) = T(\mathcal{C}_{\mathcal{R}_n}, \mathcal{V})$ (in other words, $\mathcal{C}_{\mathcal{R}_0} = \mathcal{C}_{\mathcal{R}_n}$).²

2 Summary of [4]

2.1 Basic Inference Rules for Rewriting Induction

The basic inference rules for rewriting induction of constrained TRS are proposed in [4] as follows:

Simplification

$$\frac{(\mathcal{E} \uplus \{C[l\sigma] \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E} \cup \{C[r\sigma] \approx t \llbracket \phi \rrbracket\}, \mathcal{H})}$$

where $l \rightarrow r \llbracket \psi \rrbracket \in \mathcal{R} \cup \mathcal{H}$, ϕ is satisfiable w.r.t. $\mathcal{M}$, $fv(\psi\sigma) \subseteq fv(\phi)$, and $\phi \Rightarrow \psi\sigma$ is valid w.r.t. $\mathcal{M}$.

Deletion

$$\frac{(\mathcal{E} \uplus \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E}, \mathcal{H})}$$

where $s \equiv t$ or ϕ is not satisfiable w.r.t. $\mathcal{M}$.

Expansion

$$\frac{(\mathcal{E} \uplus \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E} \cup \text{Expd}(s, t, \phi, p), \mathcal{H} \cup \{s \rightarrow t \llbracket \phi \rrbracket\})}$$

where $s \succ t$, $s \notin \mathcal{V}$, $\text{Var}(s) \supseteq \text{Var}(t) \cup fv(\phi)$, p is an R -complete position of s under ϕ ,³ and

$$\text{Expd}(s, t, \phi, p) = \{(s[r]_p)\sigma \approx t\sigma \llbracket (\phi \wedge \psi)\sigma \rrbracket \mid l \rightarrow r \llbracket \phi \rrbracket \in \mathcal{R}, \sigma \text{ is an mgu of } s|_p \text{ and } l\}$$

EQ-Deletion

$$\frac{(\mathcal{E} \uplus \{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E} \cup \{C[s_1, \dots, s_n] \approx C[t_1, \dots, t_n] \llbracket \phi \wedge \neg(\bigwedge_{i=1}^n s_i = t_i) \rrbracket\}, \mathcal{H})}$$

where $s_1, t_1, \dots, s_n, t_n \in T(\mathcal{G}, \mathcal{V})$ and $\text{Var}(s_1, t_1, \dots, s_n, t_n) \subseteq fv(\phi)$.

²This looks wrong.

³A position p of s is called *R-complete under a constraint ϕ* if for any normalized substitution σ that is ground to s , the subterm $(s\sigma)|_p$ of $s\sigma$ at position p is a redex of $\mathcal{R}$.

When we apply one of the inference rules above to $(\mathcal{E}, \mathcal{H})$ and obtain $(\mathcal{E}', \mathcal{H}')$, we write $(\mathcal{E}, \mathcal{H}) \vdash_{ri} (\mathcal{E}', \mathcal{H}')$.

Theorem 10 ([4, Theorem 4.1]) *Let $\mathcal{E}$ be a finite set of constrained equations, and $\succ$ be a reduction order with $\rightarrow_{\mathcal{R}} \subseteq \succ$. Suppose that $\mathcal{R}$ is complete and locally sound w.r.t. $\mathcal{M}$. If $(\mathcal{E}, \emptyset) \vdash_{ri} \dots \vdash_{ri} (\emptyset, \mathcal{H})$, then every equation in $\mathcal{E}$ is an inductive theorem of $\mathcal{R}$.*

To prove this theorem, we need the following propositions and lemmas.

Lemma 11 ([4, Theorem A.1.2]) *Let $\rightarrow_1$ and $\rightarrow_2$ be binary relations over A . Then, $\leftrightarrow_1^* = \leftrightarrow_2^*$ if all of the following hold:*

- $\rightarrow_1 \subseteq \rightarrow_2$,
- $\rightarrow_2$ is well-founded, and
- $\rightarrow_2 \subseteq \rightarrow_1 \circ \rightarrow_2^* \circ \leftrightarrow_1^* \circ \leftarrow_2^*$.

Lemma 12 ([4, Lemma A.1.3]) *Suppose that $\mathcal{R}$ is terminating and locally sound w.r.t. $\mathcal{M}$. Let s and t be terms, ϕ be a constraint, and σ be a substitution that is ground to s and t . If $\mathcal{M} \models \phi\sigma$, then there exists a normalized substitution θ such that*

- $s\sigma \rightarrow_{\mathcal{R}}^* s\theta$,
- $t\sigma \rightarrow_{\mathcal{R}}^* t\theta$,
- θ is ground to s and t , and
- $\mathcal{M} \models \phi\theta$.

Lemma 13 ([4, Lemma A.1.4]) *Let s and t be terms, ϕ be a constraint, p be an R -complete position of s under ϕ . Then, $s\sigma \rightarrow_{\mathcal{R}} \circ \leftrightarrow_{\text{Expd}(s,t,\phi,p)} t\sigma$ for any normalized substitution σ that is ground to s and t .*

Lemma 14 ([4, Lemma A.1.5]) *Let $s_1, t_1, \dots, s_n, t_n$ be terms in $T(\mathcal{G}, \mathcal{V})$, $C[\]$ be a context, ϕ be a constraint such that $\text{Var}(s_1, t_1, \dots, s_n, t_n) \subseteq \text{fv}(\phi)$. Suppose that $\mathcal{R}$ is complete w.r.t. $\mathcal{M}$. Then,*

$$\leftrightarrow_{\{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \mid \phi\}} \subseteq \leftrightarrow_{\{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \mid \phi \wedge (\bigwedge_{i=1}^n s_i = t_i)\}} \cup \leftrightarrow_{\mathcal{R}}^*$$

on ground terms.

Lemma 15 ([4, Lemma A.1.6]) *Suppose that $\mathcal{R}$ is complete w.r.t. $\mathcal{M}$. If $(\mathcal{E}, \mathcal{H}) \vdash_{ri} (\mathcal{E}', \mathcal{H}')$, then*

$$\leftrightarrow_{\mathcal{E}} \subseteq \rightarrow_{\mathcal{R} \cup \mathcal{H}'}^* \circ (\leftrightarrow_{\mathcal{E}'} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}'}^*$$

Lemma 16 ([4, Lemma A.1.7]) *Suppose that $\mathcal{R}$ is terminating, complete, and locally sound w.r.t. $\mathcal{M}$, and $(\mathcal{E}, \mathcal{H}) \vdash_{ri} (\mathcal{E}', \mathcal{H}')$. If*

$$\rightarrow_{\mathcal{R} \cup \mathcal{H}} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{\mathcal{R} \cup \mathcal{H}}^* \circ (\leftrightarrow_{\mathcal{E}} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}}^*$$

on ground terms, then

$$\rightarrow_{\mathcal{R} \cup \mathcal{H}'} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{\mathcal{R} \cup \mathcal{H}'}^* \circ (\leftrightarrow_{\mathcal{E}} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}'}^*$$

on ground terms.

Lemma 17 ([4, Lemma A.1.8]) *Let $\succ$ be a reduction order such that $\rightarrow_{\mathcal{R}} \subseteq \succ$. Suppose that $(\mathcal{E}, \mathcal{H}) \vdash_{ri} (\mathcal{E}', \mathcal{H}')$. If $\rightarrow_{\mathcal{H}} \subseteq \succ$, then $\rightarrow_{\mathcal{H}'} \subseteq \succ$.*

Lemma 18 ([4, Lemma A.1.9]) *Let $\succ$ be a reduction order such that $\rightarrow_{\mathcal{R}} \subseteq \succ$. Suppose that $\mathcal{R}$ is complete and locally sound w.r.t. $\mathcal{M}$, and $(\mathcal{E}, \mathcal{H}) \vdash_{ri} \dots \vdash_{ri} (\mathcal{E}', \mathcal{H}')$. Then, all of the following hold:*

- $\leftrightarrow_{\mathcal{E}} \subseteq \rightarrow_{\mathcal{R} \cup \mathcal{H}'}^* \circ (\leftrightarrow_{\mathcal{E}'} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}'}^*$ on ground terms,
- if

$$\rightarrow_{\mathcal{R} \cup \mathcal{H}} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{\mathcal{R} \cup \mathcal{H}}^* \circ (\leftrightarrow_{\mathcal{E}} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}}^*$$

on ground terms, then

$$\rightarrow_{\mathcal{R} \cup \mathcal{H}'} \subseteq \rightarrow_{\mathcal{R}} \circ \rightarrow_{\mathcal{R} \cup \mathcal{H}'}^* \circ (\leftrightarrow_{\mathcal{E}'} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}'}^*$$

on ground terms, and

- if $\rightarrow_{\mathcal{H}} \subseteq \succ$, then $\rightarrow_{\mathcal{H}'} \subseteq \succ$.

2.2 Sufficient Condition of R -Completeness

It is known that R -completeness is decidable for TRSs. On the other hand, the case of constrained TRSs is not known, while it would be undecidable in general. The following theorem shows one of sufficient conditions for R -completeness.

Theorem 19 ([4, Theorem 5.1]) *Suppose that $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G}, \mathcal{V})$.⁴ Let $s \equiv C[f(s_1, \dots, s_n)]_p$ where f is a defined symbol of $\mathcal{R}$, and $s_1, \dots, s_n \in T(\mathcal{G}, \mathcal{V})$. If $\bigwedge_{f(x_1, \dots, x_n) \rightarrow r} \llbracket \phi \rrbracket^\phi$ is valid w.r.t. $\mathcal{M}$,⁵ then position p of s is R -complete under any arbitrary constraint.*

In general, constrained TRSs obtained by transforming procedural programs satisfy the above sufficient condition for R -completeness.

2.3 Disproving Inductive Theorems

The inference rules for rewriting inductions have a function to disprove inductive theorems for sound constrained TRSs.⁶

Theorem 20 ([4, Theorem 6.1]) *Let $\mathcal{E}$ be a finite set of constrained equations. Suppose that $\mathcal{R}$ is sound w.r.t. $\mathcal{M}$, and $(\mathcal{E}, \emptyset) \vdash_{ri} (\mathcal{E}' \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})$. If $s, t \in T(\mathcal{G}, \mathcal{V})$ and $\phi \wedge \neg s = t$ is satisfiable w.r.t. $\mathcal{M}$, then $\mathcal{E}$ contains a constrained equation that is not an inductive theorem of $\mathcal{R}$.*

⁴ $NF_{\mathcal{R}}$ denotes the set of ground normal forms of $\mathcal{R}$.

⁵ $x_1, \dots, x_n$ are distinct variables.

⁶ $\mathcal{R}$ is called *sound w.r.t. $\mathcal{M}$* if for any ground interpretable $s, t \in T(\mathcal{G})$, $s \leftrightarrow_{\mathcal{R}}^* t$ implies $\mathcal{M} \models s = t$. In [4], it has been shown that $\mathcal{R}$ is sound w.r.t. $\mathcal{M}$ if $\mathcal{R}$ is confluent and locally sound w.r.t. $\mathcal{M}$.

To prove this theorem, we need the following lemmas.

Lemma 21 ([4, Lemma A.2.1]) *Let $s \simeq t \llbracket \phi \rrbracket$ be a constrained equation, and p be a position of s . Then,*

$$\leftrightarrow_{Expd(s,t,\phi,p)} \subseteq (\leftarrow_{\mathcal{R}} \circ \leftrightarrow_{\{s \simeq t \llbracket \phi \rrbracket\}}) \circ (\leftrightarrow_{\{s \simeq t \llbracket \phi \rrbracket\}} \circ \rightarrow_{\mathcal{R}})$$

on ground terms.

Lemma 22 ([4, Lemma A.2.2]) *If $(\mathcal{E}, \mathcal{H}) \vdash_{ri} (\mathcal{E}', \mathcal{H}')$, then $\leftrightarrow_{\mathcal{H}' \cup \mathcal{E}'} \subseteq \leftrightarrow_{\mathcal{R} \cup \mathcal{H} \cup \mathcal{E}}^*$ on ground terms.*

Lemma 23 ([4, Lemma A.2.3]) *If $(\mathcal{E}, \mathcal{H}) \vdash_{ri} \dots \vdash_{ri} (\mathcal{E}', \mathcal{H}')$, then $\leftrightarrow_{\mathcal{H}' \cup \mathcal{E}'} \subseteq \leftrightarrow_{\mathcal{R} \cup \mathcal{H} \cup \mathcal{E}}^*$ on ground terms.*

2.4 Relaxing Side Conditions of Inference Rules

When $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G})$ holds, the side conditions of some inference rules can be relaxed.

Simplification⁻

$$\frac{(\mathcal{E} \uplus \{C[l\sigma] \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E} \cup \{C[r\sigma] \approx t \llbracket \phi \rrbracket\}, \mathcal{H})}$$

where $l \rightarrow r \llbracket \psi \rrbracket \in \mathcal{R} \cup \mathcal{H}$, ϕ is satisfiable w.r.t. $\mathcal{M}$, and $\phi \Rightarrow \psi\sigma$ is valid w.r.t. $\mathcal{M}$.⁷

Deletion⁻

$$\frac{(\mathcal{E} \uplus \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E}, \mathcal{H})}$$

where one of the following holds:⁸

- $s \equiv t$,
- ϕ is not satisfiable w.r.t. $\mathcal{M}$, or
- there exists terms $s', t' \in T(\mathcal{G}, \mathcal{V})$ and a substitution σ such that $s \equiv s'\sigma$, $t \equiv t'\sigma$, and $s' = t'$ is valid w.r.t. $\mathcal{M}$.

EQ-Deletion⁻

$$\frac{(\mathcal{E} \uplus \{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \llbracket \phi \rrbracket\}, \mathcal{H})}{(\mathcal{E} \cup \{C[s_1, \dots, s_n] \approx C[t_1, \dots, t_n] \llbracket \phi \wedge \neg (\bigwedge_{i=1}^n s_i = t_i) \rrbracket\}, \mathcal{H})}$$

where $s_1, t_1, \dots, s_n, t_n \in T(\mathcal{G}, \mathcal{V})$.⁹

When we apply one of the original or above inference rules to $(\mathcal{E}, \mathcal{H})$ and obtain $(\mathcal{E}', \mathcal{H}')$, we write $(\mathcal{E}, \mathcal{H}) \vdash_{ri-} (\mathcal{E}', \mathcal{H}')$.

⁷ $fv(\psi\sigma) \subseteq fv(\phi)$ is removed.

⁸The third condition is added.

⁹ $Var(s_1, t_1, \dots, s_n, t_n) \subseteq fv(\phi)$ is removed.

Theorem 24 ([4, Theorem 7.1]) *Let $\mathcal{E}$ be a finite set of constrained equations, and $\succ$ be a reduction order with $\rightarrow_{\mathcal{R}} \subseteq \succ$. Suppose that $\mathcal{R}$ is complete and locally sound w.r.t. $\mathcal{M}$, and $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G})$. If $(\mathcal{E}, \emptyset) \vdash_{ri-} \dots \vdash_{ri} (\emptyset, \mathcal{H})$, then every equation in $\mathcal{E}$ is an inductive theorem of $\mathcal{R}$.*

To prove this theorem, we prepare the following lemmas.

Lemma 25 ([4, Lemma A.3.1]) *Let $s_1, t_1, \dots, s_n, t_n \in T(\mathcal{G}, \mathcal{V})$, $C[\]$ be a context, and ϕ be a constraint. Suppose that $\mathcal{R}$ is terminating, complete and locally sound w.r.t. $\mathcal{M}$, and $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G}, \mathcal{V})$. Then,*

$$\leftrightarrow_{\{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \llbracket \phi \rrbracket\}} \subseteq \left(\rightarrow_{\mathcal{R}}^* \circ \leftrightarrow_{\{C[s_1, \dots, s_n] \simeq C[t_1, \dots, t_n] \llbracket \phi \wedge \neg(\bigwedge_{i=1}^n s_i = t_i) \rrbracket\}} \circ \leftarrow_{\mathcal{R}}^* \right) \cup \leftrightarrow_{\mathcal{R}}^*$$

on ground terms.

Lemma 26 ([4, Lemma A.3.2]) *Suppose that $\mathcal{R}$ is terminating, complete and locally sound w.r.t. $\mathcal{M}$, and $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G}, \mathcal{V})$. If $(\mathcal{E}, \mathcal{H}) \vdash_{ri-} (\mathcal{E}', \mathcal{H}')$, then*

$$\leftrightarrow_{\mathcal{R}} \subseteq \rightarrow_{\mathcal{R}}^* \circ (\leftrightarrow_{\mathcal{E}'} \cup \leftrightarrow_{\mathcal{R}}^*) \circ \leftarrow_{\mathcal{R} \cup \mathcal{H}'}^*$$

on ground terms.

The following is the case of disproving.

Theorem 27 ([4, Theorem 7.2]) *Let $\mathcal{E}$ be a finite set of constrained equations. Suppose that $\mathcal{R}$ is sound w.r.t. $\mathcal{M}$, $NF_{\mathcal{R}}(\mathcal{F} \cup \mathcal{G}) \subseteq T(\mathcal{G})$, and $(\mathcal{E}, \emptyset) \vdash_{ri} (\mathcal{E}' \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})$. If $s, t \in T(\mathcal{G}, \mathcal{V})$ and $\phi \wedge \neg s = t$ is satisfiable w.r.t. $\mathcal{M}$, then $\mathcal{E}$ contains a constrained equation that is not an inductive theorem of $\mathcal{R}$.*

3 Techniques in the Prototype of crisys

Consider the case to prove that $f(\vec{x}) \approx g(\vec{x})$ is an inductive theorem of $\mathcal{R}$. The fact that this equation is an inductive theorem means that f and g are equivalent from the viewpoint of *functions*. If f and g are independent,¹⁰ then in general $f(\vec{x}) \approx g(\vec{x})$ can be oriented in both directions. When expanding $f(\vec{x}) \approx g(\vec{x})$, it is difficult (or impossible) to know which direction of orienting the equation is better. Thus, we want to try both of the cases. To this end, the prototype of the implementation divides (or apply Expansion to) processes of theorem proving as follows: in expanding $s \simeq t \llbracket \phi \rrbracket$ (at root position) that can be oriented in both directions,

- the application of Expansion divides the process $(\mathcal{E} \cup \{s \simeq t \llbracket \phi \rrbracket\}, \mathcal{H})$ into $(\mathcal{E} \cup \text{Expd}(s, t, \phi, \varepsilon), \mathcal{H} \cup \{s \rightarrow t \llbracket \phi \rrbracket\})$ and $(\mathcal{E} \cup \text{Expd}(t, s, \phi, \varepsilon), \mathcal{H} \cup \{t \rightarrow s \llbracket \phi \rrbracket\})$ if $\text{root}(s)$ and $\text{root}(t)$ do not depend on each other, and
- $s \rightarrow t \llbracket \phi \rrbracket$ is selected if $\text{root}(s)$ does not depend on $\text{root}(t)$ and $\text{root}(t)$ depends on $\text{root}(s)$.

¹⁰The definitions of f and g do not depend on each other.

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